Predictive Problem Set 8 solution

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Problem 1: Logistic Regression on Weekly Data

(a) Consider the Weekly data available in the ISLR package of R. Use the data from 1990 to 2008 as training data and the data from 2009 to 2010 as test data.

(b) Using the training data, fit a logistic regression model with Direction as the response and the five lag variables plus Volume as predictors. Identify significant predictors and interpret the coefficient with the smallest p-value.

(c) Estimate the odds of the market going "Up" at median values of the predictors.

(d) Draw the ROC curve based on the training sample and comment on model performance.

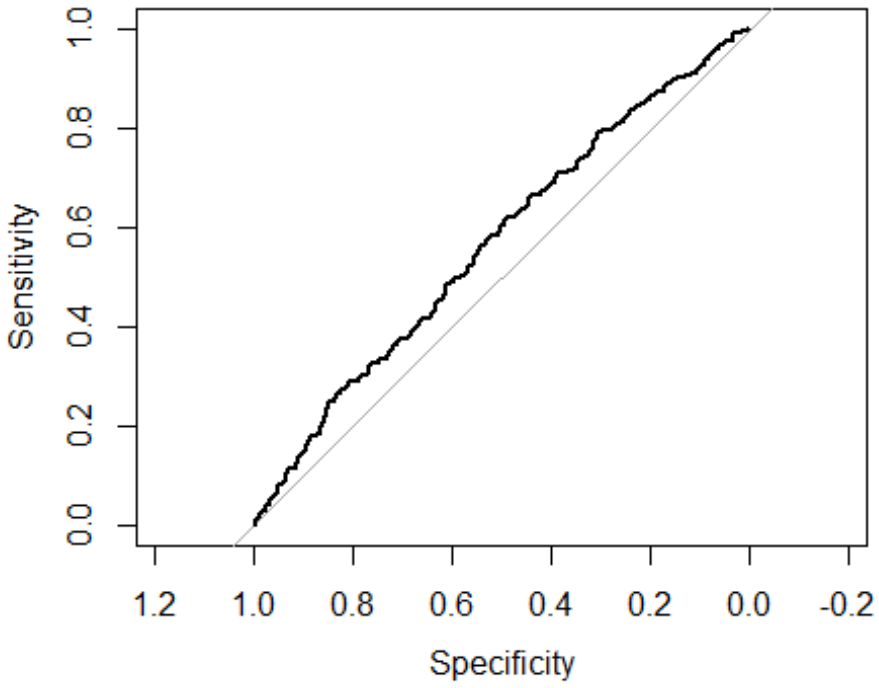
(e) Using the test data, determine the optimal cut-off that minimizes $TPR(1 - FPR)$. Report the confusion matrix and test error rate.

(f) Compare the test error rate when using only Lag_1 and Lag_2 as predictors.

```
rm(list=ls())
library(ISLR)

## Warning: package 'ISLR' was built under R version 4.3.3

data(Weekly)
train=Weekly[Weekly$Year <= 2008, ]
test=Weekly[Weekly$Year > 2008, ]
model=glm(Direction ~ Lag1 + Lag2 + Lag3 + Lag4 + Lag5 + Volume,
          data = train, family = binomial)
summary(model)

##
## Call:
## glm(formula = Direction ~ Lag1 + Lag2 + Lag3 + Lag4 + Lag5 +
##     Volume, family = binomial, data = train)
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)  0.33258    0.09421   3.530 0.000415 ***
## Lag1        -0.06231    0.02935  -2.123 0.033762 *
```

```
## Lag2         0.04468    0.02982   1.499 0.134002
## Lag3        -0.01546    0.02948  -0.524 0.599933
## Lag4        -0.03111    0.02924  -1.064 0.287241
## Lag5        -0.03775    0.02924  -1.291 0.196774
## Volume      -0.08972    0.05410  -1.658 0.097240 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 1354.7  on 984  degrees of freedom
## Residual deviance: 1342.3  on 978  degrees of freedom
## AIC: 1356.3
##
## Number of Fisher Scoring iterations: 4

# Among the predictors, Lag1 is statistically significant at the 5% level (p-value
= 0.0338).
# All other predictors (Lag2, Lag3, Lag4) are not statistically significant.
medians=apply(train[, -9], 2, median)
newdata=as.data.frame(t(medians))
prob=predict(model,newdata,type="response")
odds=prob/(1 - prob)
odds

##        1
## 1.267475

# The estimated odds of the market going up at median predictor values is 1.267
475
library(pROC)

## Warning: package 'pROC' was built under R version 4.3.3

## Type 'citation("pROC")' for a citation.

##
## Attaching package: 'pROC'

## The following objects are masked from 'package:stats':
##
##     cov, smooth, var

prob_train=predict(model, type = "response")
roc_obj=roc(train$Direction, prob_train)
```

```
## Setting levels: control = Down, case = Up

## Setting direction: controls < cases

plot(roc_obj)
```




```
auc(roc_obj)

## Area under the curve: 0.5655

# The ROC curve indicates moderate predictive ability with AUC=0.5655. The
model has limited but non-random classification power.
coords=coords(roc_obj, "best", best.method = "closest.topleft")
cutoff=coords$threshold
prob_test=predict(model, newdata = test, type = "response")
pred=ifelse(prob_test > cutoff, "Up", "Down")
table(Predicted = pred, Actual = test$Direction)

##          Actual
## Predicted Down Up
##      Down   39 55
##      Up      4  6

mean(pred != test$Direction)

## [1] 0.5673077
```

```
# True Negatives (TN) = 39
#False Negatives (FN) = 55
#False Positives (FP) = 4
#True Positives (TP) = 6
#The test error rate is 0.5673 (56.73%), indicating that the model performs poo
rly on the test data.
#The model has a strong bias towards predicting "Down", as seen from the high
number of false negatives (55). It fails to correctly identify upward movements,
suggesting weak predictive performance.
model2=glm(Direction~Lag1+Lag2,data = train, family = binomial)
prob_test2=predict(model2,newdata=test,type="response")
pred2=ifelse(prob_test2>0.5,"Up","Down")
mean(pred2!=test$Direction)

## [1] 0.4230769

#Using only Lag1 and Lag2, the test error rate is 0.4230769.This shows that a s
impler model performs better, indicating less overfitting.
```